

Tofwerk HDF5 data format

1. Introduction

HDF5 is a general purpose library and file format for storing scientific data. HDF5 was created to address the data management needs of scientists and engineers working in high performance, data intensive computing environments. As a result, the HDF5 library and format emphasize storage and I/O efficiency. HDF5 can store two primary objects: datasets and groups. A dataset is essentially a multidimensional array of data elements, and a group is a structure for organizing objects in an HDF5 file. Using these two basic objects, one can create and store almost any kind of scientific data structure, such as images, arrays of vectors, and structured and unstructured grids.

The HDF5 file format and software tools are developed by the HDF Group, a spin off from the National Center for Supercomputing Applications (NCSA) at the University of Illinois at Urbana-Champaign. For more details about the HDF5 format in general, copyright and licensing information visit the HDF Group web page (<http://www.hdfgroup.org/HDF5/>).

2. Tofwerk HDF5 basics

HDF5 has become the standard data format of the Tofwerk data acquisition software (TofDAQ) starting at version 39.8 (November 2006). The ancestor of Tofwerk HDF5 (Tofwerk binary data format v5) is no longer developed but data import and conversion (e.g. to HDF5) is supported by all new versions of TofDAQ. Tofwerk recommends all users to convert their existing binary files to HDF5 using a current version of TofDAQ in order to increase the longevity and readability of the acquired Tof-MS data.

The HDF5 in general and therefore the Tofwerk variant as well is designed to be self describing, i.e. no information other than found in the file itself is needed to interpret the datasets.

Groups are used to form logical units of different datasets and form a directory tree (similar to the directory structure of file system). Meta information is stored as attributes (name/value pairs). These attributes are always attached to the group containing the dataset(s) and not directly to the datasets.

3. HDF5 user-block

In addition to the information contained within the logical HDF5 file, data can be written at the beginning of the physical file (*user-block* in HDF terminology). The *user-block* is a fixed-length block of data located at the beginning of the file and which is ignored by the HDF5 library. Its size can be any power of 2 starting at 512 bytes. In the Tofwerk HDF5 files the user-block is plain text data and consists of a (text) header identifying the file as a Tofwerk HDF5 file and optional user supplied text which is too long for the Acquisition log (see next section) or should be readable without using a HDF5 capable reader (possible uses are lab journal, sample descriptions etc.).

Header:

```
=====
Tofwerk AG (http://www.tofwerk.com) HDF5 data file
created: 12/20/2006 5:56:16 PM
=====
```

user supplied text will appear here...

4. Groups

Fig. 1 shows the “directory” structure of a Tofwerk HDF5 file as presented by HDFView 2.3, a visual tool for browsing and editing HDF4 and HDF5 files (<http://hdfgroup.com/hdf-java-html/hdfview/>).

📁 and 📂 represent groups with and without attached attributes, 📊 stands for a dataset and 📄 represents a table (in HDF5 a table is a one-dimensional dataset of compound data type, each dataset element corresponds to a table row and each data type in the compound corresponds to a table column).

- Root group (/)
serves as an attachment point for the other main groups and has attributes attached to it that apply to more than one main group.
- AcquisitionLog group
has no attributes and contains a single table with the acquisition log.
- FullSpectra group
contains full mass spectra and has the attributes attached to it needed for the interpretation of the ToF MS data.
- RawData group
has a structure which is dependant on the DAQ hardware used for spectrum acquisition. The example shows the structure for the Acqiris AP240 using the averager firmware. No attributes are attached to the RawData group. The attributes attached to / RawData/AP240/Ch1 and .../Ch2 are needed to convert from raw data to full mass spectra.
- TimingData group
has attributes attached that describe the experiment timing and together with the BufTimes dataset allows for the reconstruction of timestamps for every saved mass spectrum.

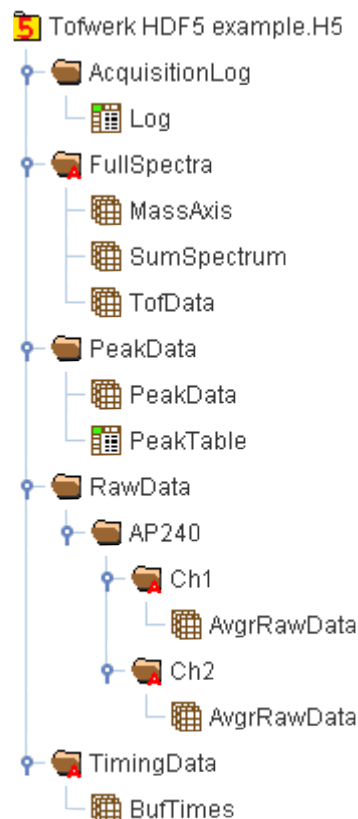


Figure 1: Structure of an exemplary Tofwerk HDF5 file

5. Attributes

Attributes are small meta data objects describing the nature and/or intended usage of a dataset or group. The attributes attached to a given group must have a unique name as they are usually addressed by name. Although the HDF5 library provides mechanisms to detect and cycle through all attributes attached to an object, the preferred way to read an attribute is to address it by its absolute name directly.

- Attributes in the root group (/)

Attribute name	Attribute type	Description
TofDAQ Version	32-bit float	TofDAQ version used for acquiring/converting the HDF5 file.
TofDAQ Build Date	string	TofDAQ build date
DAQ Hardware	string	DAQ hardware/firmware used for data acquisition
DAQ Serial	string	DAQ hardware serial number
Timing Controller	string	Timing controller used for data acquisition
Timing Controller Serial	string	Timing controller serial number

Computer ID	string	Computer specific ID
HDF5 File Creation Time	string	Date/time of acquisition/conversion (human readable)
Configuration File	string	.ini file used for acquisition
CurrentRun	32-bit int	Current run out of NbrRuns
NbrRuns	32-bit int	Total number of runs (each run generates a separate HDF5 file)
NbrSamples	32-bit int	Number of data points per mass spectrum (1 st data dimension)
NbrPeaks	32-bit int	Number of peaks defined at acquisition time
NbrWaveforms	32-bit int	Number of extractions coadded into one segment
NbrSegments	32-bit int	Number of segments (2 nd data dimension)
NbrBlocks	32-bit int	Number of blocks coadded
NbrMemories	32-bit int	Number of memories coadded
NbrBufs	32-bit int	Number of bufs (3 rd data dimension)
NbrWrites	32-bit int	Number of writes (4 th data dimension)
IonMode	string	“positive” or “negative” ions measured

- Attributes in FullSpectra group (/FullSpectra)

Attribute name	Attribute type	Description
SampleInterval	32-bit float	Sample interval in seconds
MassCalibration a	64-bit float	Conversion factors between samples (not time) and mass. The relation between <u>sample</u> index and mass is given by: $sample = a \cdot \sqrt{mass} + b$
MassCalibration b	64-bit float	
MassCalibration R0	32-bit float	Parameters related to mass resolving power. Refer to the document “Writing software for TOFs – Programmer Guide” for details.
MassCalibration m0	32-bit float	
MassCalibration dm	32-bit float	
Single Ion Signal	32-bit float	Single ion signal in mV·ns
MassCalibration nbrPoints	32-bit int	Number of mass calibration points (mass/tof sample pairs)
MassCalibration mX	64-bit float	mass / tof sample of calibration point X (X is in the range 1 to <i>MassCalibration nbrPoints</i>).
MassCalibration tX	64-bit float	

- Attributes in Ch1/2 raw data groups (/RawData/AP240/ChX)

Attribute name	Attribute type	Description
Record	32-bit int	Channel used (=1) or not (=0) for DAQ
FullScale	32-bit float	Full scale of Digitizer in Volt
Gain	32-bit float	Digitizer gain (= 5V / FullScale)
Offset	32-bit float	Digitizer baseline offset in Volt
ThresholdingEnabled	32-bit int	Hardware thresholding active (=1) or inactive (=0)
Threshold	32-bit int	Threshold level in bins (relevant only if ThresholdEnabled=1)
Baseline	32-bit int	Baseline level in bins (relevant only if ThresholdEnabled=1)
PreAmpGain	32-bit float	Gain of external preamp
DeltaPosPeak	32-bit float	Positive and negative thresholds for automatic peak detection in Volt (relevant only for advTDC firmware)

DeltaNegPeak	32-bit float	
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- Attributes in TimingData group (/TimingData)

Attribute name	Attribute type	Description
TimingMode	32-bit int	Index of timing mode used for DAQ
TofPeriod	32-bit int	Tof extraction period in ns
BlockDelay	32-bit int	Delay from block start to first DAQ trigger in ns
BlockPeriod	32-bit int	Period of 1 block in ns
PulseDelay	32-bit int	Delay from block start to emission of block pulse in ns
PulseWidth	32-bit int	Block pulse width in ns
StartDelay	32-bit int	Delay between Tof extraction pulse and DAQ trigger in ns
HV1Width	32-bit int	+ Tof extraction pulse width in ns
HV2Width	32-bit int	– Tof extraction pulse width in ns
BankSwitchPeriod	32-bit int	Memory bank switch every n extractions (SSR and aTDC only)
AcquisitionTimeZeroLo	32-bit int	Start of acquisition (number of 100-nanosecond intervals since January 1, 1601 (UTC)) FILETIME structure compatible
AcquisitionTimeZeroHi	32-bit int	

6. Datasets

In order to maximize disk I/O not all datasets shown in Fig. 1 are necessarily written to the HDF5 file. The user can choose whether or not to write the following datasets during data acquisition:

<u>HDF5 dataset(s):</u>	<u>ini file entry controlling generation of dataset:</u>
/RawData/AP240/Ch1/AvgrRawData	AutoSaveCh1
/RawData/AP240/Ch2/AvgrRawData	AutoSaveCh2
/FullSpectra/TofData	AutoSaveCombined
/PeakData/PeakData	AutoSavePeaks
/FullSpectra/MassCalibration	ReCalibration > 1
/Pressure/SensorX	IncludePressure

If the TofData dataset is present the PeakData dataset is redundant as it can easily be obtained by integrating the TofData dataset. If raw data is present, TofData and PeakData are redundant as TofData can be generated from the raw data and PeakData from the newly regenerated TofData by integration.

The Tofwerk HDF5 file format uses one-, two- and four-dimensional data arrays. As the base HDF5 libraries are written in C, the developers have adopted C's row-major order for multidimensional arrays. If you are reading HDF5 multidimensional data using the Fortran libraries or Matlab be aware that these programs use column-major order.

If you are not familiar with the terms Samples, Waveforms, Segments, Bufs, Blocks, Memories and Writes please refer to the document “Understanding TOF data”.

/AcquisitionLog/Log:

Dataset type: Table (1D array of compound data type)

of elements: variable

column name	data type	Description
timestamp	32-bit int	C timestamp in seconds since 0:00:00 January 1, 1970 UTC
timestring	string (length 26)	same time in human readable form
logtext	string (length 255)	Text describing the event

Two entries are automatically saved in the Log dataset upon data acquisition: “Acquisition started” and “Acquisition stopped”/“Acquisition aborted”.

HDF5 files generated by converting a Tofwerk binary data file to HDF5 contain entries for the actual time of data acquisition (“C:\someDir\someDataFile.bin recorded”) and for the time of conversion (“HDF5 file created”).

/FullSpectra/MassAxis:

Dataset type: 1D array of 32-bit float

of elements: NbrSamples

The MassAxis holds a mass value (amu) for every sample on the ToF dimension (1st data dimension). The dataset is redundant and saved for convenience only as the same data is easily obtained through the mass calibration parameters a and b.

/FullSpectra/SumSpectrum:

Dataset type: 1D array of 64-bit float

of elements: NbrSamples

The SumSpectrum dataset is the sum of all mass spectra in TofData. If you want an average of all spectra instead divide all elements by NbrSegments*NbrBufs*NbrWrites.

/FullSpectra/ToFData:

Dataset type: 4D array of 32-bit float

of elements: NbrWrites * NbrBufs * NbrSegments * NbrSamples

The ToFData dataset contains all recorded full mass spectra. Each mass spectrum is the sum of the three possible coaddition steps. If you want an average per extraction value, divide all elements by NbrWaveforms*NbrBlocks*NbrMemories. The unit of the intensities is mV (for the default combine mode) at the MCP signal out (before external preamps). Integration of this data leads to peak areas in mV·ns which can be divided by the single ion signal value to obtain ions.

/FullSpectra/NbrMemories:

Dataset type: 2D array of 16-bit unsigned int

of elements: NbrWrites * NbrBufs

This dataset exists only if “variable NbrMemories” feature was enabled during acquisition and contains the NbrMemories value for each recorded buf. The NbrMemories attribute attached to the root group still exists and has the initial value.

/FullSpectra/MassCalibration:

Dataset type: 2D array of 64-bit float

of elements: NbrWrites * 2

This dataset exists only if Recalibration > 1 (Automatic recalibration set to “after every write” or “after every n-th write”). For every write the mass calibration parameters a and b are stored (n times the same set for “after every n-th write”). The first entry corresponds to the the calibration parameters stored as an attribute of the /FullSpectra group. Recalibrating a file which contains this dataset will only adjust a and b stored as attributes and the values in this dataset are left untouched. Rename the dataset to some other name to use the mass calibration stored as attributes.

/PeakData/PeakData:

Dataset type: 4D array of 32-bit float

of elements: NbrWrites * NbrBufs * NbrSegments * NbrPeaks

The PeakData dataset holds the results obtained by integrating TofData. The unit of the values is ions/extraction. Multiply each element with the Tof extraction frequency ($= 1.0E9/\text{TofPeriod}$) to get a counts per second (cps) value.

/PeakData/PeakTable:

Dataset type: Table (1D array of compound data type)

of elements: variable

column name	data type	Description
label	string (length 64)	Descriptive label for the ion
mass	32-bit float	mass of the ion
lower integration limit	32-bit float	Upper and lower limit for peak integration (in mass units).
upper integration limit	32-bit float	

The PeakTable dataset is a 1:1 copy of the “Mass Table” in TofDAQ at the time of acquisition. This information is needed to (re)calculate PeakData from TofData.

/Pressure/SensorX:

Dataset type: 2D array of 32-bit float

of elements: NbrWrites * NbrBufs

This dataset exists only if IncludePressure is set. X is an integer running from 1 to the maximum of sensors that are monitored (6 for Pfeiffer MaxiGauge™). Information about the sensors are attached as attributes to the /Pressure group. All pressure values are stored in mbar (kPa).

/RawData/AP240/ChX/AvgrRawData:

Dataset type: 1D array of 32-bit int

of elements: NbrWrites * NbrBufs * NbrMemories * NbrSegments * NbrSamples

The raw data format of the AP240 ADC with averaging firmware is a 8 bit value (0-255) for every sample. The array is 1D as read from ADC card. Each mass spectrum is the sum of the three possible coaddition steps. If you want an average per extraction value, divide all elements by NbrWaveforms * NbrBlocks. Using the meta information attached to /RawData/AP240/Ch1 and /RawData/AP240/Ch2 one can convert the integer values in AvgrRawData datasets to mV and combine the two channels to the TofData dataset. For other DAQ hardware/firmware refer to the DAQ manual for a description of the raw data format.

/RawData/AP240/ChX/DescriptorLength:

Dataset type: 2D array of 64-bit unsigned int

of elements: NbrWrites * NbrBufs

This descriptor dataset stores how much raw data belongs to a given buf.

/RawData/AP240/ChX/DescriptorOffset:

Dataset type: 2D array of 64-bit unsigned int

of elements: NbrWrites * NbrBufs

This descriptor dataset stores the offset in the raw dataset for a given buf.

/TimingData/BufTimes:

Dataset type: 2D array of 64-bit float

of elements: NbrWrites * NbrBufs

The BufTimes dataset holds the start times for every buf (NbrBufs*NbrWrites). The first element should in principle be 0 but is usually on the order of 1 μ s which can also be considered to be an approximate accuracy of these values.

7. Concluding remarks

This documents describes the Tofwerk HDF5 data format as implemented at the time of writing. Any **addition** of groups, attributes or datasets (by TofDAQ or any other software) does not impair the readability of the file in a software that is not aware of these additions, they will simply be ignored. However, **changes** of groups, attributes or datasets described in this document will result in software development efforts for all software that reads Tofwerk HDF5 files and is therefore strongly discouraged.

changes from initial format definition:

additions (attributes and/or datasets) are not listed here.

v. 1.15

- the format of the PeakTable changed in version 1.15 of TofDaqRec. Before, integration boundaries were stored in samples, now they are stored in mass units (PeakTable stays valid if mass calibration changes).

v. 1.35

- raw data is no longer stored in 4D datasets. All raw datasets are 1D and contain the data and data type as read from the dataset. The DescriptorOffset and DescriptorLength datasets facilitate raw data access (only the chunk of raw data needed can be read from the file). Effective from TofDaqRec versions 1.35 and higher.

v. 1.49

- Threshold and Baseline were stored as absolute voltages before TofDaqRec 1.49. Later versions store directly the number of bins as configured by the user.

v. 1.75

- The dataset /FullSpectra/MassCalibration which is created when the “automatic recalibration” feature is enabled was a table of length NbrWrites before TofDaq 1.75. Later versions store the two mass calibration parameters a and b in a 2D dataset (64-bit floating point) of dimension 2 x NbrWrites.
- As of TofDaq 1.75 the mass calibration parameters a and b (attributes of /FullSpectra group) are stored as 64-bit floats (32-bit floats before). See also changes in the TofDaq API concerning the changes in mass calibration handling.